

12.2

Révision:

eg. $\ominus f(x,y) = e^{-(x^2+y^2)} \quad (x,y) \in \mathbb{R}^2, \quad \mathcal{B}\mathcal{D}, \quad \lambda$

positive, mesurable $\Rightarrow$ Fubini - Tonelli

$$\int_{\mathbb{R}} \int_{\mathbb{R}} e^{-(x^2+y^2)} dx dy = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-(x^2+y^2)} dy dx$$

$$\int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta \stackrel{FT}{=} \int_0^{\infty} \left(\int_0^{2\pi} e^{-r^2} r d\theta \right) dr$$

$$\stackrel{FT}{=} \int_0^{\infty} e^{-r^2} r \left(\int_0^{2\pi} 1 d\theta \right) dr$$

$$= 2\pi \int_0^{\infty} e^{-r^2} r dr$$

$$= 2\pi \left[e^{-r^2} \right]_0^{\infty} \left(-\frac{1}{2} \right) = \pi$$

$\ominus f(x,y) = \frac{\sin(xy)}{1+x^2} e^{-y^2} \quad (x,y) \in \mathbb{R}^2$

pas tj Positive $\Rightarrow$ ~~F-Tonelli~~ F-Lebesgue

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |f(x,y)| dx dy = \int_{\mathbb{R}} \int_{\mathbb{R}} \underbrace{\frac{|\sin(xy)|}{1+x^2}}_{\leq \frac{1}{1+x^2}} e^{-y^2} dx dy$$

$\ominus f(x,y) = \frac{x^2-y^2}{(x^2+y^2)^2} \quad (x,y) \in \mathbb{R}^2$

$\Rightarrow$ pas positive $\Rightarrow$ ~~FT~~

$$\left| \frac{x^2-y^2}{(x^2+y^2)^2} \right| = \left| \frac{r^2(\cos^2\theta - \sin^2\theta)}{r^4} \right| \quad \begin{matrix} x = r\cos\theta \\ y = r\sin\theta \end{matrix} = \frac{1}{r^2} |\cos 2\theta|$$

pas d'intégrabilité ~~FT~~

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \frac{x^2-y^2}{(x^2+y^2)^2} dx dy \right) = -\infty \quad \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \frac{x^2-y^2}{(x^2+y^2)^2} dy \right) dx = +\infty$$

$f(x,y)$ mesurable

positive $\xrightarrow{\text{Oui}}$ Fubini - Tonelli

$\searrow$ Non $\rightarrow$ intégrable $\xrightarrow{\text{Oui}}$ Fubini - Lebesgue
 $\searrow$ non ?

Convolution 卷积

Définition 1.5

Soit f et g deux fonctions mesurables (pour les tribus boréliennes) de $\mathbb{R}^s \rightarrow \mathbb{R}$, on appelle *produit de convolution* de f et g et on note $f * g$, l'application (lorsqu'elle a un sens)

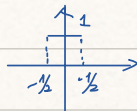
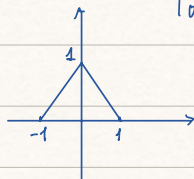
$$\begin{cases} \mathbb{R}^s \rightarrow \mathbb{R} \\ x \mapsto \int_{\mathbb{R}^s} \phi(\bullet, x) d\lambda, \end{cases}$$

où ϕ est l'application

$$(t, x) \mapsto f(t) g(x-t)$$

$$\phi: \mathbb{R}^s \rightarrow \mathbb{R} \\ t \mapsto f(t) g(x-t)$$

exemple: $f(t) = \begin{cases} t+1 & t \in [-1, 0] \\ -t+1 & t \in [0, 1] \\ 0 & \text{sinon} \end{cases}$ $g(t) = \begin{cases} 1 & t \in [-\frac{1}{2}, \frac{1}{2}] \\ 0 & \text{sinon} \end{cases}$



$$f * g(x) = \int_{\mathbb{R}} f(t) g(x-t) dt$$

changement de variable $u = x-t$

$$\int_{-\infty}^{+\infty} f(x-u) g(u) du = (g * f)$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x-u) du$$

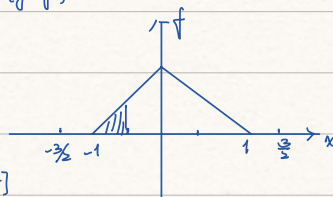
$$\stackrel{\text{COV}}{=} \int_{x-\frac{1}{2}}^{x+\frac{1}{2}} f(y) dy$$

$$x \in [-\frac{3}{2}, \frac{3}{2}]$$

$$kf = [-1, 1]$$

$$kg = [-\frac{1}{2}, \frac{1}{2}]$$

$$k_{f * g} = [-\frac{3}{2}, \frac{3}{2}]$$



propriétés:

Propriétés:

$$① f * g = g * f$$

$$② f * (g * h) = (f * g) * h$$

$$③ f * (g + h) = f * g + f * h$$

$$④ a(f * g) = (af) * g = f * (ag)$$

$$\cancel{⑤ (f * g)' = f' * g + f * g' = f' * g = f * g'}$$

$$⑥ \overline{f * g} = \overline{f} * \overline{g}$$

$$⑦ f * G(t) = \overline{f} * g(t)$$

$$= \int_{\mathbb{R}^s} f * g(z) dz$$

$$\int_{\mathbb{R}^s} f(t) \int_{\mathbb{R}^s} g(y) h(x-t-y) dy dt$$

$$= \int_{\mathbb{R}^s} h(x-t) \int_{\mathbb{R}^s} f(y) g(x-t-y) dy dt$$

tels que, $\forall x \in \mathbb{R}^S, |f(x+\eta) - f(x)| < \varepsilon$ (car continue)

$$(x, y) \in (\mathbb{R}^S)^2$$

$$|f * g(x) - f * g(y)| = \left| \int_{\mathbb{R}^S} f(t) g(x-t) dt - \int_{\mathbb{R}^S} f(t) g(y-t) dt \right|$$

linéarité
symétrique

$$= \left| \int_{\mathbb{R}^S} (f(x-t) - f(y-t)) g(t) dt \right|$$

$$|x-y| = \eta$$

$$\leq \int_{\mathbb{R}^S} |f(x-t) - f(y-t)| |g(t)| dt$$

inégalité
de Hölder

$$\leq \|f(x-t) - f(y-t)\|_q \|g\|_p \quad \frac{1}{p} + \frac{1}{q} = 1$$

$$p = 1$$

$$\sum \|g\|_1$$

$$p = +\infty$$

$$\sum (\lambda(k) + \eta) \|g\|_\infty$$

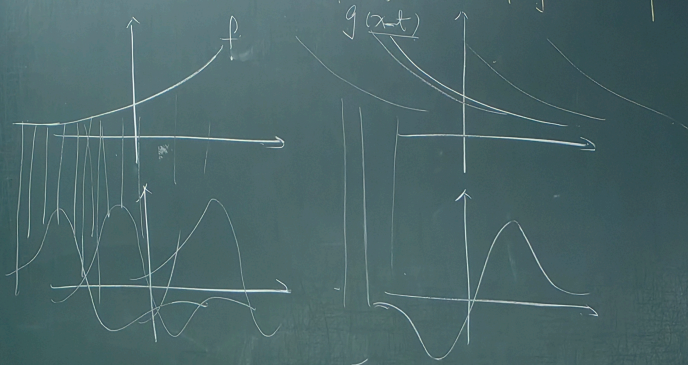
$$p \in]1, +\infty[, \frac{1}{p} + \frac{1}{q} = 1$$

$$\|f(x-t) - f(y-t)\|_q \|g\|_p$$

Donc, $f * g$ est uniformément continue $\sum (\lambda(k) + \eta)^{\frac{1}{q}} \|g\|_p$

② $f \in L^p$ ($p > 1, \frac{1}{p} + \frac{1}{q} = 1$), l'application $g \mapsto f * g$ est Lipschitzienne sur L^q

③ $f \in L^p, g \in L^q$ ($\frac{1}{p} + \frac{1}{q} = 1$), $f * g$ est uniformément continue



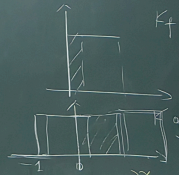
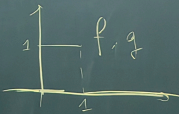
Example

① $f = \mathbb{1}_{[0,1]}$

$g = \mathbb{1}_{[0,1]}$

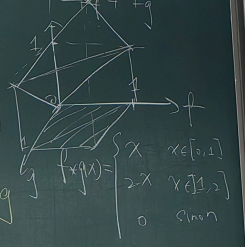
$f * g$

$K_{f+g} = [0, 2]$



$K_f = [0, 1]$

$K_{g(x)} = [-1, 1]$

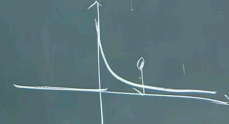
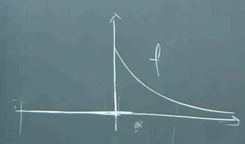


③ $f(x) = e^{-x} \mathbb{1}_{x \geq 0}$

$g(x) = e^{-2x} \mathbb{1}_{x \geq 0}$

$f * g$

$f * g(x) = \begin{cases} x & x \in [0, 1] \\ 2-x & x \in [1, 2] \\ 0 & \text{sinon} \end{cases}$



$K_{f+g} = [0, +\infty[= K_f + K_g$

$K_f = [0, +\infty[$

$K_g = [0, +\infty[$

$$f * g = \int_{-\infty}^{+\infty} f(t) g(x-t) dt$$
$$= \int_{-\infty}^{+\infty} (e^{-t} \mathbb{1}_{t \geq 0}) \times (e^{-2(x-t)} \mathbb{1}_{x-t \geq 0}) dt$$

③ $\mathcal{F}(f * g) = \mathcal{F}(f) \times \mathcal{F}(g) \Rightarrow \text{pas p\'eriodique} \rightarrow \text{pas p\'eriodique}$